Design and Simulation of a Low-Cost Programmable Electronic Load for Power Electronics Laboratories

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***Abstract*—Power electronics laboratories require test loads for safely and accurately testing rectifiers, single-phase AC–AC and AC–DC converter circuits, batteries, and switching power supplies. Most technical educational facilities, however, use simple resistor networks of fixed-value resistors to emulate DC current loads. Such fixed resistor bank test loads offer only a single current level for a given supply voltage and provide no inherent protection for expensive laboratory equipment under test. This paper presents a simple, low-cost, analog Programmable Electronic Load (PEL) together with its Proteus simulation results, designed for safely testing a variety of undergraduate laboratory power supplies. The load circuit consists of an NE555 timer integrated circuit configured as a 48 kHz PWM oscillator, an LM358 dual operational amplifier serving as an overcurrent protection comparator and a closed-loop current-control error amplifier, and a logic-level power MOSFET (IRLZ44N) driving a buck-type inductor/diode current-sink circuit. A precision 0.1 Ω sense resistor provides current feedback. A rotary switch and two potentiometers enable selection and fine adjustment of discrete DC current setpoints drawn from the supply under test. Green and red LEDs are provided to indicate normal operation and overcurrent fault conditions, respectively. The design does not require a microprocessor, has a minimal component count, and is fully compatible with existing student laboratory equipment. Detailed circuit operation, a block-level description, and closed-loop simulation results obtained using Proteus 8.17 SP5 are presented, together with virtual oscilloscope waveforms. The simulation results confirm a stable, current-regulated output, correct MOSFET switching behaviour, and accurate overcurrent protection across a range of operating scenarios. The proposed PEL is contrasted with fixed-resistor approaches and with recently reported analog and digital electronic load implementations in the literature.**

***Index Terms*—Programmable electronic load, NE555, LM358, buck converter, logic-level MOSFET, power electronics.**

I. Introduction

Programmable Electronic Load (PEL) emulators are widely used to replicate controllable DC load conditions for testing and evaluating power supplies, batteries, fuel cells, and DC–DC converters in industrial and research settings [1], [2]. Unlike a fixed resistor, which draws a current proportional to the applied voltage, a PEL can be configured to operate in constant-current (CC), constant-voltage (CV), constant-resistance (CR), or constant-power (CP) mode, enabling repeatable tests across varying DC voltage levels [3]. Commercial programmable DC loads offer high power capability, multiple operating modes, high bandwidth, and digital communication interfaces such as USB, LAN, and GPIB [4], [5]. However, these instruments are prohibitively expensive for undergraduate teaching laboratories and conceal the underlying control loop mechanisms from students [6], [7].

Discrete resistor banks are widely used in engineering laboratories, particularly in India, to test rectifiers and power converters [8], [9]. A significant disadvantage is the complete absence of protection for the equipment under test; laboratory kits are frequently damaged by short circuits or incorrect wiring. A low-cost PEL can be integrated into existing laboratory setups to enable safe, repeatable experiments and to illustrate fundamental closed-loop control concepts such as PWM control, current sensing, and feedback [2], [10]. There is a growing interest in the academic and research community in developing low-cost DC active load solutions [2], [10], [11]. Most solutions reported in the literature, however, require microcontrollers with complex firmware or are of industrial grade [3], [11].

There is therefore a clear requirement for a simple PEL module that can be retrofitted into existing fixed-resistor-based power electronics experiment setups to demonstrate closed-loop current control using components covered in standard undergraduate curricula. This paper addresses this gap through three principal contributions. First, a compact six-block PEL architecture is presented, covering circuit topology, control technology, and user interface. Second, all circuit components are described in detail and the complete design is verified through Proteus simulation. Third, the proposed PEL is compared with recent analog and digital programmable electronic load implementations published after 2020 [2], [11]–[13].

The remainder of this paper is organised as follows. Section II reviews relevant commercial and academic DC electronic load implementations. Section III describes the proposed PEL architecture and detailed circuit design. Section IV presents simulation probe readings and oscilloscope waveform results. Section V identifies directions for future development, and Section VI concludes the paper.

II. Literature Review

A. Commercial Programmable DC Loads

Modern commercial programmable DC electronic loads are designed for high-power applications, offering wide voltage and current ranges, multiple operating modes (CC, CV, CP, and dynamic ramp), and high-fidelity transient simulation capability [1], [4], [5]. Intended for R&D, production testing, and EV battery validation, contemporary instruments approach or exceed kilowatt and megawatt power levels with very high slew rates [6], [7]. They incorporate high-resolution user interfaces and standard digital communication interfaces, together with built-in data logging [1]. Representative commercial products spanning across a broad power range are documented in [14]–[17]. Owing to their complexity and cost, these instruments are not suitable for basic undergraduate teaching laboratories, and their internal control loop implementations remain hidden to students [8], [9].

B. Low-Cost and Educational DC Active Loads

To reduce costs for battery capacity testing and general laboratory use, several low-cost DC active load modules have been developed [10], [18]. Huynh *et al.* [10] present an open-source hardware DC load module for battery capacity evaluation, featuring modular power management and a freely available bill of materials. Santoro *et al.* [2] propose a cost-efficient DC active load based on a buck–boost-derived converter topology, published in *Frontiers in Energy Research*, demonstrating its application for battery test and measurement in a university laboratory setting. A number of do-it-yourself (DIY) electronic load designs have been published on hobbyist platforms such as Seeed Studio [18] and educational online channels [19]. Although these designs demonstrate inexpensive construction paths, they generally lack a rigorous analysis a of closed-loop system dynamics. In practice, safety and protection issues are often not formally addressed, and several important design parameters remain unspecified.

C. Microcontroller-Based Programmable Loads

A number of publications report microcontroller-based programmable DC loads with multiple operating modes and digital control [3], [11]. Afsal *et al.* [11] detail an ESP32-based programmable electronic load supporting CC, CR, and CP modes using a PI control structure with ten-turn potentiometers for entry of setpoint. Overcurrent protection, thermal shutdown, and under-voltage lockout are implemented. The power stage uses MOSFET switching with Hall-effect current sensors, test sequences and profiles are stored in the flash memory. Peng *et al.* [3] present a programmable DC load incorporating keypad and LCD interfaces for adjustment of voltage and current setpoint. Additional reported implementations [20] describe embedded microcontroller designs with keypad/LCD interfaces. However, such designs require substantial firmware development that falls outside the scope of a purely analog design exercise.

D. Advanced Control of DC Electronic Loads

Advanced control strategies for DC electronic loads have been studied in recent literature. Jiang and He [12] propose a self-tuning PID controller which is based on particle swarm optimization (PSO), showing improved dynamic response and reduced overshoot over a conventional fixed-parameter PID through MATLAB simulation. He and Jiang [13] subsequently report a DC electronic load using a BP neural-network-based PID strategy, achieving further gains in steady-state regulation and transient performance. Although these approaches highlight the value of advanced control for load accuracy, their computational demands place them well beyond the reach of a low-cost analog implementation.

E. MOSFET Gate Driving and Converter Topologies

Advanced MOSFET gate-drive techniques and optimized buck converter designs for programmable electronic load power stages have been studied in recent literature. Beohar *et al.* [21] present a digitally controlled synchronous buck DC/DC converter with automatic PFM–PWM mode transition, which is aimed at maintaining high efficiency over a wide load range. Hamdy *et al.* [22] examine MOSFET gate driving using bootstrap circuits, with a focus on enhancement of dynamic load continuity. Du *et al.* [23] propose a digital closed-loop active gate driver for static and dynamic current sharing of paralleled SiC MOSFETs. Earlier work by Seixas [24] demonstrated a 300 A dynamic electronic load based on a modified interleaved buck+boost converter, establishing the viability of switch-mode converter topologies for high-current active load applications. Although these techniques yield high performance, they are not required for the low-power PEL proposed in this work, which employs a simpler NE555-based analog PWM approach.

F. Summary and Identified Gap

Recent literature encompasses high-power commercial programmable loads, low-cost active load modules for battery and general-purpose laboratory use, microcontroller-controlled

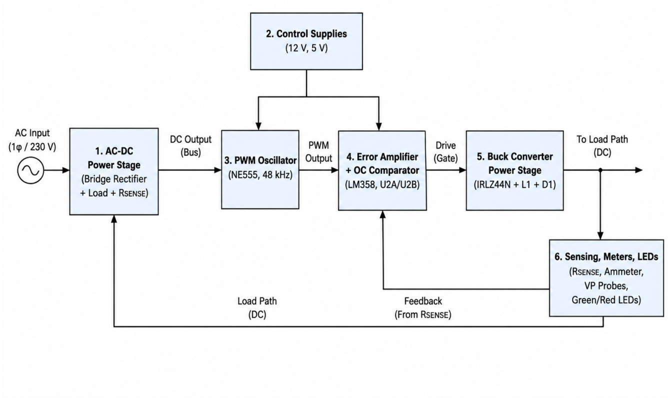

Fig. 1: Proposed PEL Block Diagram organized into six functional blocks.

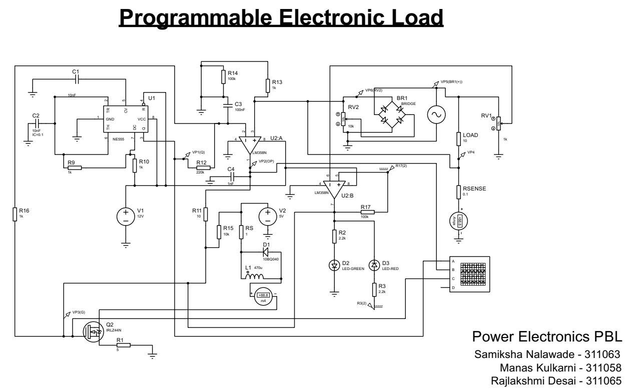

Fig. 2: Complete circuit implementation of the proposed programmable electronic load.

multi-mode programmable loads, and advanced control strategies for high-performance power conversion [7], [10]–[12], [23]. A clear gap exists for a simple, fully analog, low component-count PEL that can be integrated into existing laboratory kits using only components covered in typical undergraduate Power Electronics and Drives (PED) courses. The proposed PEL implements closed-loop constant-current regulation with current sensing and hardware overcurrent protection, all of which can be readily explained and demonstrated in an undergraduate laboratory context.

III. Proposed Methodology

A. Overall Architecture and Block Diagram

The complete PEL circuit is illustrated in The complete PEL circuit is illustrated in Fig. 1, divided into six functional blocks: (1) AC–DC power supply, (2) control supply, (3) PWM oscillator, (4) error amplifier and overcurrent protection, (5) buck converter power stage, and (6) sensing, display, and status indication.

The AC–DC power stage accepts the laboratory AC supply. Full-bridge rectifier BR1 converts AC to pulsating DC, which constitutes the main current path through the $10\,\Omega$ load resistor (LOAD) and the parallel $0.1\,\Omega$ sense resistor R_{SENSE}.

The circuit runs from two separate DC supply rails. $V_1 = 12\,\text{V}$ is used for the NE555 and LM358 control circuitry, and $V_2 = 5\,\text{V}$ powers the buck converter and sensing sections. This keeps switching noise out of the control loop and away from the feedback signal.

The PWM oscillator is implemented with an NE555 timer in astable mode, providing a 48 kHz carrier at Pin 3. Timer components are $R_9 = 1\,\text{k}\Omega$, $R_{10} = 1\,\text{k}\Omega$, and $C_2 = 10\,\text{nF}$.

The LM358N dual op-amp is used with U2A as the current-regulation error amplifier and U2B as the overcurrent comparator. The overcurrent threshold is set via potentiometer RV2.

The power stage is a buck converter built around logic-level MOSFET IRLZ44N (M1), inductor $L_1 = 470\,\mu\text{H}$, and Schottky diode D1 (10BQ040). Load current is regulated through the MOSFET gate voltage, which is driven by the error amplifier output.

Load current feedback is derived from R_{SENSE}, which produces a proportional sense voltage. Simulation probes VP1–VP6, an ammeter, and a three-channel virtual oscilloscope in Proteus allow real-time waveform monitoring. Status LEDs D2 (green, normal operation) and D3 (red, overcurrent fault) complete the user interface.

The complete circuit implementation of the proposed programmable electronic load is shown in Fig. 2.

B. NE555 PWM Oscillator

The NE555 is operated in free-running astable mode. Timing capacitor C_2 charges through R_9 and R_{10} and discharges through R_{10} alone. The free-running oscillation frequency is given by

$$f = \frac{1.44}{(R_9 + 2R_{10})\,C_2} \tag{1}$$

Substituting $R_9 = R_{10} = 1\,\text{k}\Omega$ and $C_2 = 10\,\text{nF}$ into (1) yields

$$f = \frac{1.44}{(1\,\text{k}\Omega + 2 \times 1\,\text{k}\Omega) \times 10\,\text{nF}} \approx 48\,\text{kHz} \tag{2}$$

With these component values the duty cycle is approximately 66.7% and the Pin 3 output exhibits a peak-to-peak amplitude of approximately 11 V, as confirmed by simulation probe VP1 and oscilloscope Channel A. The NE555 output is attenuated and coupled to the inverting input of error amplifier U2A through $R_{12} = 220\,\text{k}\Omega$.

C. Closed-Loop Current Regulation Using LM358 (U2A)

Load current is measured using shunt resistor $R_{\text{SENSE}} = 0.1\,\Omega$. The resulting sense voltage is

$$V_{\text{sense}} = I_{\text{load}} \times R_{\text{SENSE}} = I_{\text{load}} \times 0.1\,\Omega \tag{3}$$

V_{sense} (monitored at probe VP4) is applied to the non-inverting input (Pin 3) of error amplifier U2A. The attenuated NE555 PWM signal at Pin 2 of U2A provides the current reference. The error amplifier output at Pin 1 (VP2) drives the MOSFET gate via series resistor $R_{16} = 1\,\text{k}\Omega$; pull-down resistor $R_{15} = 10\,\text{k}\Omega$ ensures a clean turn-off of M1.

If V_{sense} falls below the setpoint, the error amplifier output increases, raising the gate voltage and increasing MOSFET conduction. Conversely, if V_{sense} exceeds the setpoint, the output decreases, reducing conduction. Feedback resistor $R_{14} = 100\,\text{k}\Omega$ and filter capacitors $C_3 = 100\,\text{nF}$ and $C_4 = 1\,\text{nF}$ stabilise the control loop and prevent op-amp saturation.

D. Overcurrent Protection Using LM358 (U2B) and RV2

The second section of the LM358, U2B, is configured as a comparator with hysteresis for hardware overcurrent protection. Its non-inverting input (Pin 5) monitors V_{sense}, while its inverting input (Pin 6) is connected to the wiper of potentiometer RV2, which sets the trip threshold between V_{CC} and ground. When V_{sense} exceeds the threshold voltage, the U2B output (Pin 7) goes HIGH, activating the red LED D3 through $R_3 = 2.2\,\text{k}\Omega$. Hysteresis resistor $R_{17} = 100\,\text{k}\Omega$ prevents chattering around the trip point. The overcurrent trip current is approximated by

$$I_{\text{trip}} = \frac{V_{\text{threshold}}}{R_{\text{SENSE}}} = \frac{V_{\text{threshold}}}{0.1\,\Omega} \tag{4}$$

enabling straightforward adjustment of the protection threshold for different experiment requirements by varying RV2.

E. Buck Converter Power Stage and MOSFET Selection

The IRLZ44N is a logic-level N-channel MOSFET with a gate threshold voltage of 1–2 V and full enhancement at 4–5 V gate drive [25]. This enables direct gate drive from the NE555/LM358 output voltages without an additional gate-drive integrated circuit. Its drain–source voltage rating of 55 V and high continuous drain current capability are well suited to the voltage and current levels encountered in undergraduate power electronics experiments.

When M1 turns on, current builds up through L_1, storing energy in the inductor's magnetic field. When M1 turns off, that stored energy continues to drive current through Schottky diode D1 (10BQ040), keeping load current nearly constant. A standard silicon diode would suffer from high reverse recovery losses at the 48 kHz switching frequency; the near-zero reverse recovery time of the 10BQ040 Schottky makes it well suited to this application [25].

IV. Results and Discussion

A. Simulation Probe Readings

Table I lists the key Proteus simulation probe readings recorded under steady-state conditions at a representative operating point.

The NE555 oscillates at the frequency given by (2). The error amplifier U2A output has reached a near-constant DC level, which confirms stable closed-loop current regulation. A MOSFET gate voltage of approximately 3.05 V exceeds the IRLZ44N logic-level threshold, which in turn confirms active conduction. The inductor branch current of 88.8 mA is consistent with low-power buck-type active load operation.

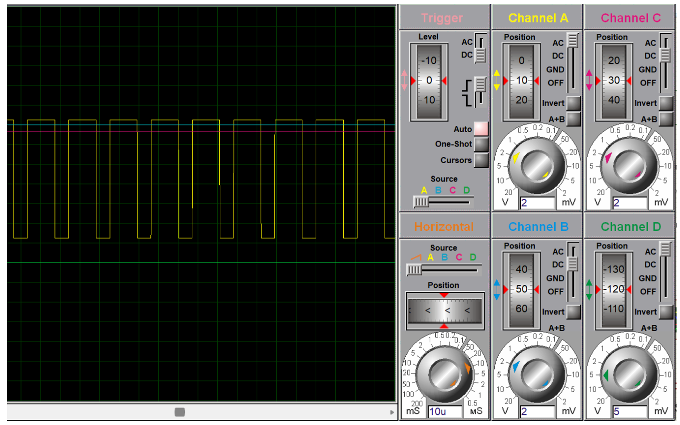

Fig. 3: Simulated oscilloscope waveforms: (A) 48 kHz PWM output at NE555 Pin 3 (VP1), (B) error amplifier output (VP2), (C) MOSFET gate voltage (VP3).

B. Oscilloscope Waveform Analysis

The simulated oscilloscope waveforms are shown in Fig. 3. At a timebase of 50 ms/div, the 48 kHz PWM waveform at Channel A appears as a solid block, since more than 2400 switching cycles fall within each 50 ms division. Reducing the timebase to 20 μs/div resolves individual square pulses with a period of 20.8 μs, corresponding to 48 kHz, which is consistent with the theoretical value of (2). Channel B displays the near-constant DC output of error amplifier U2A, confirming stable closed-loop operation. Channel C closely follows the error amplifier output through the gate resistor network, confirming the expected MOSFET gate drive waveform. All results are in agreement with the theoretical analysis of Section III.

C. Design Verification and Performance Assessment

The proposed PEL satisfies three key design requirements. *Topology*: The PEL incorporates a full-bridge rectifier and a buck-type DC–DC converter as its power stages (Fig. **??**), both of which are fundamental industrial circuit configurations taught in undergraduate Power Electronics and Drives courses [2], [3].

Control technology: Closed-loop PWM current control is realised using an NE555 oscillator as the carrier generator and an LM358 error amplifier, with R_{SENSE} providing current feedback, as described in Sections IIIC–D. The resulting control voltage can be observed at VP2, with the corresponding MOSFET gate drive at VP3 (Table I).

User interface: The desired current level is selected via potentiometer RV1 and the overcurrent protection threshold is adjusted using RV2. The green and red LEDs give a clear indication of normal and fault operating conditions, respectively.

D. Qualitative Comparison with Existing Methods

The proposed PEL is more flexible than a fixed resistor bank and is considerably simpler and cheaper than commercial programmable loads or recently reported microcontroller-based designs [4], [11]. Unlike the ESP32-based load of [11],

TABLE I: Key Simulation Probe Readings Under Steady-State Conditions

Probe / Instrument	**Observed Value (Typical)**	**Interpretation**
VP1 (NE555 Pin 3)	11.99 V to 2.67 V (switching)	48 kHz PWM oscillation confirmed per (2)
VP2 (U2A output)	≈2.82 V (near-DC)	Error amplifier holding control voltage for MOSFET gate
VP3 (MOSFET gate)	≈3.05 V	Gate voltage above IRLZ44N threshold; MOSFET conducting
VP4 (LOAD–R_{SENSE})	Small mV-level signal	Corresponds to low main-path current at this operating point
VP5 (BR1 output)	Positive DC level	Correct AC–DC rectification by full-bridge BR1
VP6 (RV2 wiper)	Setting dependent	Adjustable overcurrent threshold per (4)
Ammeter (L_1 branch)	88.8 mA	Average inductor current in buck converter branch

the proposed design requires no microcontroller, firmware, analogue-to-digital converter, or digital communication interface; it nonetheless implements PWM generation, current sensing, and error correction through analog circuitry alone. PSO-optimized PID [12] and neural-network-based PID [13] approaches deliver better transient performance, but add a level of computational complexity that is neither needed nor suitable for a low-cost analog design.

Quantitative benchmarking against commercial or microcontroller-based loads was not included in this study. The present implementation demonstrates improved flexibility over fixed resistors, provides overcurrent fault indication, and exposes the closed-loop control structure that is not accessible in commercial black-box instruments. Future hardware realizations will enable direct performance comparison in terms of regulation accuracy, bandwidth, and transient response against standard laboratory bench loads.

V. Future Scope

The present work employs Proteus simulation at power levels which is representative of a typical undergraduate laboratory experiment. Multiple avenues for further development are identified.

Building the circuit on a PCB with adequate heat sinking and thermal management would allow measurement of efficiency, temperature rise, and reliability in the long-term. Extension to CR and CP operating modes can be achieved using analog or simple microcontroller front-end circuitry, while retaining the same power stage. Integrating low-cost measurement integrated circuits can support digital display of voltage, current, and power.

Thermal protection through temperature sensing and fan speed control can be incorporated, which is common in modern commercial programmable loads [26]. For more advanced exercises, the analog control loop can serve as a platform to compare a classical PI controller against PSO-optimized PI [12] or neural-network-enhanced PID [13], without changing the power stage and sensing circuitry.

The design can also be extended to implement AC active loads or bidirectional active loads for inverter and regenerative drive testing, building on recent work in reconfigurable electronic load emulators [2], [25]. Higher-power versions of the PEL can be benchmarked against existing research-grade active DC loads to evaluate efficiency and transient response under laboratory conditions that are realistic.

VI. Conclusion

This study describes the design, circuit analysis, and Proteus simulation of a low-cost, fully analog Programmable Electronic Load (PEL) for power electronics laboratories. The design employs an NE555 timer as a 48 kHz PWM oscillator (Section IIIB), an LM358 dual operational amplifier for closed-loop current regulation, hardware overcurrent protection (Sections IIIC–D), and an IRLZ44N logic-level MOSFET driving a buck-type inductor/diode current-sink circuit (Section IIIE).

Simulation results verify correct 48 kHz PWM generation, stable error amplifier operation, appropriate MOSFET gate drive, and a regulated buck-converter inductor branch current of 88.8 mA, as listed in Table I and shown in Fig. 3. The PEL provides a significant flexibility than a fixed resistor load, enables programmable current setpoints, and provides hardware overcurrent fault indication for safe use of laboratory equipment.

Unlike the microcontroller-based and advanced-control DC electronic loads found in the literature [11]–[13], the proposed design takes a different approach. It avoids digital control entirely, and still clearly demonstrates the principles of closed-loop feedback, current sensing, and power MOSFET switching through standard, cost-effective analog components. All functional blocks are directly accessible and pedagogically transparent. Future work will address hardware realization, efficiency measurement, and extension to additional operating modes.

References

[1] ITECH Electronics, "IT8500+ series programmable DC electronic load," Product Datasheet, 2026, accessed: 2026. [Online]. Available: https://www.itechate.com/en/product/dc-electronic-load/IT8500plus.html

[2] D. Santoro, N. Delmonte, P. Cova, and A. Toscani, "A cost-efficient DC active load laboratory solution," *Frontiers in Energy Research*, vol. 10, p. 1078010, Dec. 2022. [Online]. Available: https://www.frontiersin.org/journals/energy-research/articles/10.3389/fenrg.2022.1078010/full

[3] J. Peng, Y. Chen, Y. Fang, and S. Jia, "Design of programmable DC electronic load," in *Proc. International Conference on Industrial Informatics – Computing Technology, Intelligent Technology, Industrial Information Integration (ICIICII)*, 2016, pp. 351–355.

[4] Chroma ATE Inc., “Programmable DC electronic load model 63200A series datasheet,” Product Datasheet, Rev. Aug. 2023, 2023. [Online]. Available: https://assets-us-01.kc-usercontent.com/ecb176a6-5a2e-0000-8943-84491e5fc8d1/f5cb41cb-2f11-4ec9-b33b-6197e9ae97b8/63200A-EN.pdf
[5] TDK-Lambda, “SFL multi-functional programmable DC electronic load: Fast response and linear turn-on,” Application Note, Sep. 2020. [Online]. Available: https://www.sunforcetech.com/2020/10/multi-functional-programmable-dc-electronic-load-offers-fast-response-and-linear-turn-on-with-multiple-operating-load-modes-in-300w-and-1kw-power-ratings/
[6] SkyQuest Technology, “Programmable DC electronic loads market size and growth report 2032,” Market Research Report, Jan. 2025. [Online]. Available: https://www.skyquestt.com/report/programmable-electronic-loads-market
[7] The Insight Partners, “Programmable DC electronic loads market report 2031,” Market Research Report, Mar. 2026. [Online]. Available: https://www.theinsightpartners.com/reports/programmable-dc-electronic-loads-market
[8] Scientech Technologies, “Programmable DC electronic loads — product overview,” Product Page, Oct. 2024. [Online]. Available: https://scientechworld.com/dc-electronics-loads/
[9] Scientific Mes-Technik, “Programmable DC electronic loads — DCL series,” Product Page, 2026, accessed: 2026. [Online]. Available: https://www.scientificindia.com/product-details/programmable-dc-electronic-load-DCL-Series
[10] M. N. Huynh, Q. M. Lam, C. T. Truong, H. H. Nguyen, and V. T. Duong, “Low-cost electronic DC load module design for battery capacity evaluation,” *HardwareX*, vol. 23, p. e00679, Jul. 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2468067225000574
[11] A. A. Afsal, F. M. A. K. Ahammed, V. K. Vishnu, and E. Paul, “Microcontroller based programmable load,” *International Journal of Innovative Research in Technology*, vol. 12, no. 10, pp. 8238–8246, Feb. 2026. [Online]. Available: https://ijirt.org/article?manuscript=193410
[12] L. Jiang and K. He, “Research on DC electronic load system based on PSO-PID algorithm,” *Journal of Power Electronics and Power Systems*, vol. 1, no. 1, pp. 1–10, Nov. 2022. [Online]. Available: https://www.pioneerpublisher.com/jpeps/article/download/20/5
[13] K. He and L. Jiang, “Research on DC electronic load system based on BP neural network PID control,” *Academic Journal of Engineering and Technology Science*, vol. 5, no. 12, pp. 56–61, Dec. 2022. [Online]. Available: https://francis-press.com/papers/8386
[14] PPST Solutions, “DC electronic loads — product family,” Product Page, Sep. 2025. [Online]. Available: https://ieeexplore.ieee.org/document/9487365
[15] Chroma Systems Solutions, “DC loads archives — ultra-low voltage DC load 0.25 V at 2000 A,” Product Page, Sep. 2024. [Online]. Available: https://asu.elsevierpure.com/en/publications/a-digitally-controlled-dc-dc-buck-converter-with-automatic-digita/
[16] ET System Electronic, “N69200 series DC electronic load datasheet,” Product Datasheet, 2021. [Online]. Available: https://www.et-system.de/fileadmin/user_upload/website/Datenblaetter/DC/Datasheet_N69200_Series_DC_Electronic_Load.pdf
[17] Scientific Mes-Technik, “AC + DC electronic load,” Product Page, 2026, accessed: 2026. [Online]. Available: https://www.scientificindia.com
[18] Seeed Studio, “Basic electronics: Electronic load and battery testing,” Online Article, May 2020. [Online]. Available: https://www.seeedstudio.com/blog/2020/05/26/basic-electronics-electronic-load-and-battery-testing/
[19] Various Authors, “DIY electronic load for battery, charger, and solar panel testing,” Online Video Tutorial, Feb. 2025. [Online]. Available: https://www.youtube.com
[20] J. Peng and C. Chen, “Design of programmable DC electronic load: Programmable power, resistance and current,” Academic Project Report, Shiv Nadar University, Aug. 2022. [Online]. Available: https://www.academia.edu/30378853
[21] N. Beohar, D. Mandal, V. Parasuram, A. M. Reddy, A. Kumar, V. N. K. Malladi, K. Pappu, C. Fu, P. C. Adell, and B. Bakkaloglu, “A digitally controlled DC-DC buck converter with automatic digital PFM to PWM transition scheme,” in *Proc. IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2021, pp. 517–522.
[22] A. Hamdy, S. El-Sayed, and R. Amr, “Performance of MOSFET driven via a bootstrap capacitor for dynamic load continuity enhancement,” *Journal of Electrical and Computer Engineering*, 2022. [Online]. Available: https://onlinelibrary.wiley.com/doi/epdf/10.1155/2022/2273819
[23] L. Du, X. Du, S. Zhao, Y. Wei, Z. Yang, L. Ding, and H. A. Mantooth, “Digital closed-loop active gate driver for static and dynamic current sharing of paralleled SiC MOSFETs,” *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 12, no. 2, pp. 1372–1384, 2023. [Online]. Available: https://par.nsf.gov/servlets/purl/10492914
[24] C. Seixas, “A 300 A dynamic electronic load based on a modified buck+boost interleaved converter,” *Eletrônica de Potência*, 2006.
[25] EDN Network, “Design a 100 A active load to test power supplies,” Online Technical Article, Dec. 2020. [Online]. Available: https://www.edn.com/design-a-100a-active-load-to-test-power-supplies/
[26] Vitrek, “Achieving accurate battery testing with DC electronic loads,” Application Note, Sep. 2025. [Online]. Available: https://vitrek.com/achieving-accurate-battery-testing-with-dc-electronic-loads/